JAYDIP NAKARANI

Software Engineer

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EXPERIENCE (8+ Years)

Freelancer (July 2017 – Nov 2020)

- Developed strong backend skills with Python and web scraping using BeautifulSoup and Selenium, focusing on data extraction and transformation.
- Built REST APIs using Django and Flask and integrated PostgreSQL and MySQL databases for data storage and optimization.
- Practiced Test-Driven Development (TDD) and Agile practices, collaborating with clients through feedback cycles and directly troubleshooting legacy systems.
- Managed small projects, enhancing skills in requirement gathering, client communication, and decision-making.

Sr. Python Developer

GlobaliaSoft LLP (Dec 2020 – Jan 2022)

- Led a small Python team, set coding standards, and mentored junior developers across backend and automation projects.
- Built scalable backend systems using Django, DRF, and MongoDB for data-driven client applications. Developed smart scraping pipelines using Scrapy, Selenium, and proxy rotation to handle large and complex websites.
- Integrated basic ML/NLP workflows for data cleaning, text extraction, and rule-based classification.
- Worked with AWS and Docker to deploy and optimize high-volume data services.
- Enhanced web scraping capabilities for high-volume applications with Scrapy and proxy management.
- Strengthened experience in cloud deployments on AWS with Docker, optimizing infrastructure for data-heavy applications.

Sr. Python AI-ML Developer

BitCoding Solutions (Feb 2022 – Present)

- Worked as a technical lead on AI/ML projects, designing agent systems, multimodal pipelines, and document-processing workflows in Python.
- Built multi-agent architectures using OpenAI, Claude, LangGraph, and MCP with continuous training, testing, and retraining loops.
- Developed ML/DL pipelines including YOLOv8 training, PDF/QR extraction, voice-cloning workflows, and protocol-to-video generation.
- Designed and delivered backend APIs with FastAPI for KYC verification, medical video generation, invoice processing, and agent orchestration.
- Integrated cloud services (AWS EC2, S3, RDS) and deployed containerized applications using Docker.
- Led end-to-end development of real-time avatar systems using STT, TTS, lip-sync models, and gesture animation pipelines.
- Managed project planning, client discussions, and delivery timelines to ensure smooth execution and results.

SKILLSET

Backend: Python, Django, Django REST Framework (DRF), FastAPI, Flask, ORM, GraphQL, PostgreSQL, MySQL, MongoDB, Redis, Celery, Microservices Architecture
Frontend: JavaScript, ReactJS, jQuery, React Redux, HTML5, CSS3, SCSS, Bootstrap, DOM Manipulation
Web Scraping: Scrapy, Requests, Selenium, Puppeteer, BeautifulSoup (bs4), LXML, XPath, AWS S3, RDS, Lambda, Proxy Management, CAPTCHA Bypass, Data Parsing
Machine Learning & AI: TensorFlow, PyTorch, Hugging Face Transformers, Scikit-Learn, OpenAI GPT, NLP (Natural Language Processing), Rasa, Data Preprocessing, Feature Engineering
Data Engineering: ETL Pipelines, Data Cleaning, Data Visualization, SQL, NoSQL, Data Warehousing, Data Analysis
Cloud & DevOps: AWS (EC2, S3, RDS, Lambda), Google Cloud Platform (GCP), Docker, Kubernetes, CI/CD Pipelines
Version Control & Collaboration Tools: Git, GitLab, Jira, Asana
Other: Data Structures (stack, queue, graph, linked list), TDD (Test-Driven Development), API Documentation (Swagger, Postman), Debugging, Performance Optimization

Certifications

- **Amazon Web Services (AWS):** Cloud Essentials, Architecting, File Storage, Serverless, Compute, Migration Foundations
- **IBM Watson and Machine Learning**: Technical Essentials, Machine Learning Essentials
- **Remote Work Professional Certification (RWPC™)**

EDUCATION

Parul University, Vadodara - B. Tech (CSE)
Aug 2016 - Apr 2020

- Acquired a solid foundation in **programming languages** including **JavaScript, Python, and C++**, with a focus on writing efficient, clean code.
- Developed a strong understanding of **data structures** and **algorithms**, including time and space complexity, for optimized programming solutions.
- Gained hands-on experience with **MVC frameworks**, emphasizing organized code architecture and design patterns for scalable applications.
- Engaged in projects and coursework that applied theoretical knowledge to practical programming tasks, building skills essential for real-world problem-solving.

PROJECTS

Agentic System (Mar 2025– Present) Python, FastAPI, OpenAI, Claude, LangGraph, MCP, Agent Frameworks

- Designed and implemented a full multi-agent automation system using LangGraph as the meta-layer orchestrator.
- Built an onboarding agent that interacts with businesses, collects requirements, and identifies the needed agent workflows.
- Developed a design agent responsible for creating blueprints, deciding the agent structure, number of agents, and flow.

- Implemented a build agent that constructs agents based on requirements and prepares them for training and testing.
- Created a training and testing pipeline that evaluates agent accuracy, performance, and overall behavior.
- Generated performance reports and built a mechanism to detect low-performing agents automatically. Developed an auto-retrain pipeline to improve agent accuracy by updating knowledge and re-training weak agents.
- Built a department-level agent ("Caio") that manages team-level interactions, KPIs, sub-agent tasks, and responses.
- Integrated MCP for tool calling, enabling agents to use tools dynamically during workflows.
- Ensured smooth agent-to-agent communication for complex multi-step tasks.
  Developed internal pipelines for testing, training, and continuous improvement of all agents.

KYC Verification System for Payment App (Oct 2024 – Mar 2025)

Python, FastAPI, ReactJS, YOLOv8, Docker, AWS EC2/S3, PostgreSQL, RDS, Pytest

- Built an AI-driven KYC verification system to automate Aadhaar and PAN card validation for payment account onboarding.
- Prepared and cleaned a high-quality dataset of PAN and Aadhaar documents for model training. Annotated the dataset using Roboflow to ensure accurate document and field detection.
- Fine-tuned YOLOv8 to extract key information from uploaded documents.
- Evaluated model accuracy and performance using a robust test dataset.
- Developed FastAPI-based REST APIs for document upload, verification, and data extraction.
- Integrated frontend (ReactJS) with backend APIs to provide a smooth user verification flow.
- Dockerized the entire application and deployed it on AWS EC2, using S3 for file storage.
- Implemented Pytest test cases to ensure API reliability and model performance.
- Extracted essential fields including Name, DOB, Mobile Number, Aadhaar Number, PAN Number, Signature, and Address.

Videoreach – Medical Protocol to Voice + Video Generation (Feb 2024 – Sept 2024)

Python, FastAPI, Pydantic AI, Claude API, LlamaParse, Eleven Labs, Logfire

- Built an agent pipeline that extracts medical protocol text (endoscopy, colonoscopy, etc.) using LlamaParse.
- Implemented a validation layer to ensure extraction confidence before generating instructions. Created a script generation agent that converts protocols into step-wise guidance (diet, timing, medicines).
- Separated scripts into episodes for conversational and instructional flow.
- Integrated Eleven Labs for doctor-voice cloning and high-quality text-to-speech output.
- Built a video integration agent that synchronizes audio and video, generating final instructional videos for patients.
- Developed a patient interaction agent that sends video/script for confirmation and handles patient replies.
- Used Logfire for monitoring agent actions, errors, and system events.
  Delivered a complete multimodal workflow generating both video + audio outputs from medical protocols.

Intelligent Invoice Processing System (Aug 2023 – Jan 2024)

Python, FastAPI, Claude 3.5, Sonnet, PDF Extraction, QR Code Decoding

- Developed an AI-based system to extract fields from digital and scanned invoices with high accuracy.
- Implemented QR code extraction and decoding from invoices to support government-linked verification.
- Built a validation agent to verify extracted fields, including cross-checking with government portals.
- Converted extracted and validated fields into clean, structured JSON.
- Added error tracking, audit logs, and monitoring for all extraction and validation steps.
- Ensured the system works for multiple invoice formats and variations across vendors.

Real-Time Hyper Realistic Talking Avatar (May 2023 – July 2023)

Python, ML, OpenAI STT/TTS, StyleGAN, SMPL, MuseV, MuseTalk, MusePose, Docker, AWS (ECS), FAST API

- Assessed and strategized the integration of MuseV, MuseTalk, and MusePose models.
- Ensured the cohesive operation of these models to produce realistic talking avatars.
- Integrated MuseV for visual rendering, providing high-quality visual representations.
- Implemented MuseTalk for text-to-speech (TTS) functionality and lip-sync capabilities, ensuring accurate and synchronized speech movements.
- Utilized MusePose for realistic gesture animations, enhancing the lifelike quality of the avatars.
- Developed and optimized a prototype for real-time text-to-speech with synchronized animations.
- Ensured the system's responsiveness and accuracy in real-time applications.
- Implemented OpenAI's speech-to-text (STT) and AI agent for conversational abilities, enabling interactive and responsive avatars.
- Conducted thorough testing to gather feedback and identify areas for improvement.
- Replaced initial OpenAI systems with custom TTS, STT, and AI agents to tailor the solution to specific project requirements.
- Enhanced the system's performance and flexibility by leveraging custom-built models.
- Deployed the application on AWS using Docker, ensuring scalability and reliability.
- Conducted final testing phases to validate the system's functionality and performance under various conditions.
- Integrated StyleGAN for generating realistic facial expressions, adding depth and expressiveness to the avatars.
- Utilized CMU's MoCap dataset to achieve highly realistic movement and behavior patterns, further enhancing the avatar's realism.

Vendor Product Data Management System (Jan 2023 – Apr 2023)

Python, ML, NLP, Fast API, AWS, GitLab, Textract, SQL

- Streamlining vendor product management by creating a cohesive environment for seamless data extraction, validation, and attribute updates from diverse structured (Excel, CSV) and unstructured (PDF) documents within vendor catalogs.
- Crafted a specialized model with a header classification system for precise data extraction. Implemented a data validation layer for accuracy.
- Developed a multi-layered model, including a PDF preprocessing parser and a conversion model, for transforming unstructured PDF data into a structured format.

- Introduced NLP techniques and an ensemble ML model to identify product-related tags, enhancing the efficiency of data extraction from both Excel catalogs and PDFs.
- Implemented a specialized pipeline for managing and updating prices and handling changes in product attributes, ensuring real-time database accuracy.
- Seamlessly orchestrated the entire workflow, covering dynamic vendor catalog handling, data extraction, and storage. Ensured efficient and accurate handling of diverse data formats and complexities in vendor catalogs.

Car damage detection & QR code and signature extraction (Sept 2022 – Dec 2022)

Python, ML, DL, Fast API, docker, Postgres, AWS, GitLab,

- **Car Damage Detection:**

- Annotate images using Roboflow to label different types of car damage.
- Train a car damage detection model using Detectron2 library.
- Download the trained model and deploy it to a Fast API framework.
- Develop an API endpoint that accepts car images and returns the detected damage details.
- Integrate the model with a PostgreSQL database to store results and metadata.

- **QR Code Extraction:**

- Use the pyzbar library to extract QR codes from both images and PDFs.
- Implement functionality for extracting QR codes from single files or entire folders.

- **Signature Extraction:**

- Utilize OpenCV library to extract signatures from various documents.
- Develop robust methods to handle different signature styles and complexities.

StepGuide (Feb 2022 – Aug 2022)

HTML, CSS, JavaScript, Django(backend), GCP, PostgreSQL, GitHub

- Designed and implemented UI, backend and database from scratch
- Understood the requirements thoroughly
- Implemented authentication and authorization mechanisms to ensure secure access to the application.
- Optimized database queries and overall system performance to enhance the application's responsiveness.
- Investigated and resolved bugs and issues reported by users or identified during testing.
- Integrated third-party APIs or services for additional functionality, such as payment processing or analytics.
- Reported the regular status to the higher authorities on time.
- Implemented CI/CD Pipeline for deployment.

OpenHouseDirect (June 2021 – Jan 2022)

Python, Django, Django rest framework, AngularJS, PostgreSQL, GCP, Celery, Elastic search

- A platform where people can **collaborate with real-estate property in Arizona, US**. Users can sell, buy or lease their property with our platform
- Have integrated google cloud publisher-subscriber to achieve message queuing to implement syndication process of **ARMLS** to platform database
- Have implemented **Oauth2** to implement authentication process for users and also implemented SSO with google and Facebook
- Have designed quite an enhanced website with material-UI using **angular JS**
- Have deployed web application on google cloud platform app version and used google cloud SQL for database

Carinfo (Dec 2020 – May 2021)

Python, Flask, Flask-SQL Alchemy, Docker, AWS, EC2, HTML 5, Bootstrap, CSS, Redis

- An online marketplace to maintain all car-related stuff online like search for new/old cars, add to the garage, upload vehicle history, etc.
- Have implemented E-signature for vehicle document and handled media/file with AWS S3 storage
- Have provided a robust solution for design with nicer UI/UX for user experience
- Have designed architecture for DB and migrated over AWS RDS instance and implemented complex database structure with salable solution
- Authentication handled by JWT authentication and session handled with cookies
- Implemented Redis to manage a cache of web application

G4+ (Dec 2019 – Nov 2020)

Python, Django Rest Framework, AWS SES, SNS, Elemental Media converter, Cognito (User pool, Identity pool), RDS, S3, Lambda, Cloud front, server less architecture, Pytest, mux, Redis, Kafka

- Combination of social media applications (Instagram, TikTok, Zoom, Facebook)
- Have provided technical analysis to manage user trafficking with backend server
- Integrated AWS services to enhance the performance of the backend with server less architecture
- Optimized database query with Django ORM
- Implemented live streaming video using the mux platform
- Integrated Agora API to achieve live video call features
- Provided robust solutions of rest APIs to enhance performance
- Architect database design with the following ACID properties
- Design micro services to distribute server load
- Implemented server architecture for Kafka service for meeting schedule and notify end users

Warehouse Management System (Nov 2018 – Dec 2019)

Python (Django/Flask), OIDC, Elastic Search, RabbitMQ, Azure, k8s, Data Lake

- Documented technical designs, architecture decisions, and troubleshooting procedures for future reference and knowledge sharing
- Optimized database queries and performance of the microservice to ensure scalability and efficiency.
- Implemented security measures such as encryption and access control to protect sensitive financial data.
- Evaluated and integrated third-party libraries, APIs, or services to enhance functionality and accelerate development.
- Written test cases for the functionalities
- Participated in sprint planning, backlog grooming, and daily stand-up meetings to provide input on technical feasibility, estimation, and prioritization of tasks.
- Reported the regular status to the higher authorities on time.
- Contributed to the continuous improvement of development processes and tools, advocating for automation, CI/CD pipelines, and DevOps practices.

Web Scrapper (July 2017 – Nov 2018)

Python, Request, Scrappy, Beautiful Soup, LXML, Selenium, Pandas, Cloud flare

- Developed and executed web scraping solutions across various domains, including **Real Estate**, **Sales Leads** (Instagram, Facebook, TikTok), **Finance**, **Travel**, and **E-commerce** (Amazon, Alibaba).
- Structured scraped data into formats like **TXT, PDF, JSON, API, CSV, XML, XLS, and SQL**, ensuring compatibility for data analysis and reporting.

- Automated data delivery pipelines to **AWS**, **Dropbox**, **AWS S3**, and **Oracle**, facilitating seamless integration with data storage platforms.
- Designed a **master-slave architecture** to scrape millions of records using **AWS Lambda** with **proxy rotation** to bypass Cloudflare, optimizing for high-volume data collection.
- Implemented Selenium scripts to automate data extraction from secured banking sites, retrieving and structuring real transaction data.
- Completed high-scale projects, including:
  1. **Google Maps Scraping** (location-based data)
  2. **Bing Scraping** (with email extraction)
  3. **Twitter & YouTube Scraping** (posts, comments, hashtags)
  4. **Amazon & eBay Scraping** (product details)
  5. **Reddit & Yelp Scraping** (posts, reviews)
  6. **Twilio, NumberBarn, and Bandwidth** (company data)
  7. **Instagram Email Scraping** (1M+ business emails for profiles with 10k+ followers)

PERSONAL DETAILS

Hobbies Learning new skills, reading, outdoor sports

Languages Known: English, Gujarati, Hindi

Personal Assets: Quick learner, strong communicator, analytical thinker, problem-solver, enthusiastic about continuous learning, team player, and self-motivated.

SUMMARY AND PROFILE

- Seeking a role where I can actively contribute to the organization's success while expanding my skills and expertise in a dynamic environment.
- A fast learner who adapts quickly to new technologies, I am passionate about continuous growth and ready to evolve with project needs, preferring a flexible approach over focusing on a single technology.

Thank you.